Unforced Navier–Stokes

Compact symmetric stresses, exact coefficient margins, and formal verification on the three-torus

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Abstract

The absence of external forcing makes the three-dimensional Navier–Stokes regularity problem substantially more restrictive than a construction with prescribed smooth forcing. We study exact mathematical prerequisites for a candidate unforced finite-time singularity on the periodic three-torus. An explicit compact symmetric-divergence inverse is proved for smooth axisymmetric meridional sources supported in an annulus, with vanishing total axial force as the necessary and sufficient compatibility condition. A separate two-diagonal pressure-neutral construction exhibits a sharp radial sign obstruction. Finite seven-ray stress algebra yields the exact coefficient margin 15971/122900 and sharp componentwise amplification factor 14. A compact source-matched dual test excludes a strong physical entrywise stress bound at the corresponding numerical scale. For the frozen cylindrical seven-ray frame, an exact matrix reconstruction identifies that bound with a sufficient four-coordinate H8 stress estimate. We give the exact three-inequality description and sharp dual support function of the original seven-coefficient cone. An exact rational Bernstein certificate shows that even this larger absolute coefficient budget fails for the specified manufactured meridional source. The larger, one-sided seven-ray positivity cone has a distinct asymmetric sharp dual support function. An exact rational enclosure shows that the same dual field does not obstruct this larger cone; no inverse is thereby constructed. A kernel-checked matrix identity then transfers the exact positive-cone dual to any rotating source frame, yielding a necessary integrated bound without assuming the frame is constant. None of these diagnostic tests decides the authentic H8 residual. We also derive the exact pressure-compensated annular flux identity and its two sharp frame-dependent cone inequalities. The pressure boundary trace is essential: an axial source primitive alone cannot be bounded by the stress cone when that trace is free. The source-prescribed separable shell shaper also admits an exact quadratic flux reduction, including its often-leading viscous shear. An endpoint uniqueness lemma rules out an $O(b^2)$ stress-shear budget for any nonzero compact cutoff with a nondegenerate viscous second-jet coefficient, provided the remaining terms have at most first-jet size. The hypothesis is not established for the complete H8 residual. Normalized quarter-turn localization requires twelve ordered nonlinear overlap terms; an explicit local construction shows that these terms can carry three quarters of the core kinetic shear. A separately chosen smooth annular cutoff and normalized toroidal pressure illustrate the remaining source choices, and a combined rational example checks the full flux ledger against direct stress integration. These are test functions, not the original unfrozen H8 cutoff or pressure. For a core localization sharing the shaper cutoff, a cylindrical circulation identity reduces cancellation of the viscous second jet to an exact axial-core moment and one reference-circle trace. Lean additionally checks the meridional pointwise inverse and both integral antiderivative identities, alongside the finite cone algebra, physical residual expansion, and conditional terminal-speed retention. These results do not yet construct the source-matched, scale-sharp, positive-cone-compatible compact stress or a genuine zero-force singular solution. The global unforced existence question remains open.

1 The problem and the mathematical objective

We work at viscosity one on the unit three-torus $\mathbb{T}^3 = \mathbb{R}^3/\mathbb{Z}^3$. For a smooth velocity $u(t,x) \in \mathbb{R}^3$ and pressure $p(t,x) \in \mathbb{R}$, the unforced incompressible equations are

$$\partial_t u + (u \cdot \nabla)u - \Delta u + \nabla p = 0, \qquad \operatorname{div} u = 0. \tag{1}$$

The specific target is to find nonzero smooth divergence-free periodic initial data and a solution smooth for $0 \le t < 1$, of finite kinetic energy on a fundamental cell, whose speed becomes unbounded as $t \uparrow 1$. The source-free condition in (1) is essential: a verified construction with a nonzero smooth force does not establish this target. The Clay problem statement makes this distinction explicit [1]. The published OpenAI Lean formalization states Navier–Stokes breakdown with smooth forcing; its unforced finite-time result concerns Euler rather than Navier–Stokes [2].

Our approach isolates small results that are meaningful independently of the unresolved global theorem. First, a source must admit a compact symmetric-divergence correction. Second, if that correction must remain in a prescribed positive stress cone, its components need quantitative control relative to a density envelope. Third, all terms in the actual nonlinear physical residual must cancel with a single periodic pressure. Finally, a singular leading trace can survive only if the correction is provably subleading. Sections 2–7 establish exact parts of this chain.

The construction of compact symmetric anti-divergence operators under force and torque compatibility is not new; see, for example, the support-preserving operator and its explicit moment correction in [3, §3.6 and Appendix A]. The direct axisymmetric formula below is offered as an elementary source-specific coordinate realization, not as a priority claim for compact symmetric anti-divergence. The new obstacle for the present programme is the additional shrinking-support, weighted-seven-ray positivity requirement, which is not supplied by general existence of a compact tensor. Section 5 also shows that a natural *stronger* physical-component envelope is incompatible with one explicit source, even modulo pressure gradients. We subsequently exclude the original absolute coefficient envelope for the same manufactured source by a sharper dual calculation; we then determine the sharp support of the distinct one-sided cone and test the same source against it. The exact frame pullback supplies the correct dual coordinates for the genuine spatially varying source. The actual H8 source remains a separate problem.

2 A compact symmetric inverse in an annular region

Let $0 < a < b$ and $c < d$, and write $\Omega = (a,b) \times (c,d)$ in meridional cylindrical coordinates (r,z). All functions in this section are independent of the azimuthal angle θ. The orthonormal cylindrical frame is (e_r, e_θ, e_z). For a symmetric tensor with components

$$S = K\, e_r \otimes e_r + L\, e_\theta \otimes e_\theta + M(e_r \otimes e_z + e_z \otimes e_r) + C\, e_z \otimes e_z, \tag{2}$$

standard cylindrical differentiation gives

$$(\operatorname{div} S)_r = \partial_r K + \frac{K - L}{r} + \partial_z M, \tag{3}$$

$$(\operatorname{div} S)_z = \partial_r M + \frac{M}{r} + \partial_z C, \tag{4}$$

$$(\operatorname{div} S)_\theta = 0.$$

The $1/r$ terms are geometric, not lower-order terms that may be discarded near a shrinking cylindrical support.

Theorem 2.1 (Explicit compact meridional inversion)**.** *Let $f_r, f_z \in C_c^\infty(\Omega)$, and extend $f = f_r e_r + f_z e_z$ by zero to $\mathbb{R}^3$ in cylindrical coordinates. There is a symmetric $S \in C_c^\infty(\mathbb{R}^3; \mathrm{Sym}_3)$, axisymmetric and supported in the same open annular cylinder, with $\operatorname{div} S = f$, if and only if*

$$\int_a^b \int_c^d r f_z(r,z)\,\mathrm{d}z\,\mathrm{d}r = 0. \tag{5}$$

The construction uses no nonlocal spatial projection, but it asserts no positive-semidefinite or weighted-cone bound on S.

Proof. For necessity, integrate the axial component of $\operatorname{div} S = f$ against the cylindrical Jacobian r. Compact support and (4) give

$$\int_a^b \int_c^d r f_z\,\mathrm{d}z\,\mathrm{d}r = \int_a^b \int_c^d \left[\partial_r(rM) + r\partial_z C\right]\mathrm{d}z\,\mathrm{d}r = 0.$$

For sufficiency, define the axial source marginal and its radial primitive by

$$J(r) = \int_c^d f_z(r,z)\,\mathrm{d}z, \qquad H(r) = \frac{1}{r}\int_a^r sJ(s)\,\mathrm{d}s. \tag{6}$$

Since $r > 0$ on $[a,b]$, H is smooth. The vanishing moment (5) makes H identically zero outside a compact subinterval of (a,b). Choose any $\eta \in C_c^\infty((c,d))$ normalized by $\int_c^d \eta(z)\,\mathrm{d}z = 1$, and set

$$M(r,z) = H(r)\eta(z). \tag{7}$$

The identity $(rH)' = rJ$ implies

$$\int_c^d \left(f_z - \partial_r M - \frac{M}{r}\right)\mathrm{d}z = J - H' - \frac{H}{r} = 0.$$

Consequently the axial primitive

$$C(r,z) = \int_c^z \left[f_z(r,\zeta) - \partial_r M(r,\zeta) - \frac{M(r,\zeta)}{r}\right]\mathrm{d}\zeta \tag{8}$$

vanishes both below and above the axial support; it is smooth and compactly supported in Ω. Choose any $K \in C_c^\infty(\Omega)$, including $K = 0$, and define

$$L = K + r\big(\partial_r K + \partial_z M - f_r\big). \tag{9}$$

Each component of (2) is smooth and compactly supported away from the axis. Substitution in (3) gives

$$\partial_r K + \frac{K - L}{r} + \partial_z M = \partial_r K - (\partial_r K + \partial_z M - f_r) + \partial_z M = f_r,$$

and (8) gives $\partial_r M + M/r + \partial_z C = f_z$. Thus $\operatorname{div} S = f$ as claimed. □

Remark 2.2 (Scope of the construction)**.** The theorem concerns a meridional source with $f_\theta = 0$ and support strictly separated from $r = 0$. An azimuthal source introduces a different angular-momentum compatibility condition, and extending the construction across the axis requires separate regularity conditions on the cylindrical components. The tensor in the theorem may be indefinite, and its size can deteriorate when the support shrinks. None of these omissions is harmless for the weighted H8 attachment problem.

3 Pressure-neutral gauges and a sign obstruction

The mixed and hoop degrees of freedom in Theorem 2.1 are not optional decorations. A restricted diagonal ansatz exhibits a genuine obstruction to maintaining one sign.

Proposition 3.1 (Two-diagonal gauge)**.** *Let $F \in C_c^\infty(\Omega)$, put $K = rF_r$ and $C = F + rF_r$, and take $S = Ke_r \otimes e_r + Ce_z \otimes e_z$. Then*

$$\operatorname{div} S = \nabla C. \tag{10}$$

In particular its divergence is a pressure gradient, so it cannot by itself represent a prescribed source with a nonzero solenoidal part.

Proof. The radial divergence is $K_r + K/r$. Since $K = rF_r$, $C_r = F_r + K_r = K/r + K_r$. The axial divergence is C_z, which proves (10). □

Proposition 3.2 (Restricted radial sign condition)**.** *If K and C are compactly supported in $r \in (a, b)$ at a fixed z and obey $C_r = K_r + K/r$, then*

$$\int_a^b K(r, z)/r \, \mathrm{d}r = 0. \tag{11}$$

Consequently, $K \geq 0$ forces $K \equiv 0$ for this restricted two-diagonal class.

Proof. Integrate the derivative identity between a and b; the endpoint terms of K and C vanish by compact support. Since $r > 0$, continuity and (11) imply that a nonnegative K is identically zero. □

This conclusion does not exclude a sign-changing correction around a positive background tensor or a compact construction with nonzero hoop and mixed components. Indeed, Theorem 2.1 explicitly uses both of those components. The more difficult question is whether one can choose them while retaining a prescribed strict coefficient cone with uniform shrinking-support estimates.

4 The exact seven-ray coefficient margin

The finite cone appearing in the current H8 source has seven rays. With aperture $t = 1/4$ they are

$$\begin{aligned} &v_1 = (1, 0, 0), \quad v_2 = (1, t, 0), \quad v_3 = (1, -t, 0), \\ &v_4 = (1, 0, t), \quad v_5 = (1, 0, -t), \\ &v_6 = (1, t, t), \quad v_7 = (1, t, -t). \end{aligned} \tag{12}$$

Their normalized positive coefficient centre is $c_i^{(0)} = 236/1229$ for $i = 3$ and $172/1229$ for every other index. The raw source covariance and its trace-free rank-one identity are established by exact finite arithmetic and in `UnforcedPatching.SevenRayCenter.lean`.

Write $(A, B, C, D) \in \mathbb{R}^4$ for the four trace-free stress coordinates. The resulting coefficient perturbations, in ray order, are

$$\delta c = \left(-\tfrac{3}{2}D,\ 2C - 4D,\ -2C - 4D,\ 2A - 8B + 4D,\ -2A + 8B + 4D,\ 8B,\ -8B\right). \tag{13}$$

Proposition 4.1 (Sharp componentwise cone estimate)**.** *For $m = \max(|A|, |B|, |C|, |D|)$ the map (13) satisfies*

$$\max_{1 \leq i \leq 7} |\delta c_i| \leq 14m, \tag{14}$$

and the number 14 *is optimal. If* $\rho \geq 0$ *and*

$$m \leq \frac{15971}{1720600}\rho, \tag{15}$$

then for every ray $c_i^{(0)}\rho + \delta c_i \geq \rho/100$.

Proof. The absolute row sums in (13) are respectively 3/2, 6, 6, 14, 14, 8, 8; hence the operator norm from $\ell^\infty(\mathbb{R}^4)$ to $\ell^\infty(\mathbb{R}^7)$ is at most 14. At $(A, B, C, D) = (1, -1, 0, 1)$, the fourth coordinate equals 14, so the constant is sharp. Since

$$\frac{172}{1229} - \frac{1}{100} = \frac{15971}{122900},$$

inequality (15) implies $|\delta c_i| \leq 14m \leq (15971/122900)\rho$. As $c_i^{(0)} \geq 172/1229$, the claimed floor follows. □

The seven-ray claim is a finite linear-algebra statement. It provides the exact smallness target for a potential compact correction but gives no estimate for the particular tensor delivered by Theorem 2.1. In particular, no implication from unweighted compact solvability to (15) is available.

Proposition 4.2 (The frozen-frame physical tensor map)**.** *In the local co-rotating frame* $(e_1, e_2, e_3) = (e_r, e_\theta, e_z)$ *of the seven-ray construction, let* $G = \sum_{i=1}^7 \delta c_i v_i \otimes v_i$. *The exact physical tensor perturbation and its trace-free pressure representative are*

$$G = \begin{pmatrix} -\frac{3}{2}D & C & A \\ C & -\frac{1}{2}D & B \\ A & B & \frac{1}{2}D \end{pmatrix}, \quad H = G + \frac{D}{2}I = \begin{pmatrix} -D & C & A \\ C & 0 & B \\ A & B & D \end{pmatrix}. \tag{16}$$

In particular, if $m = \max(|A|, |B|, |C|, |D|)$, *then* $\max_{i,j} |H_{ij}| = m$ *in that frame. Conversely, the coordinates are recovered as* $A = H_{rz}$, $B = H_{\theta z}$, $C = H_{r\theta}$, *and* $D = H_{zz}$. *The image is the four-dimensional subspace of symmetric trace-free matrices with* $H_{\theta\theta} = 0$ *(hence* $H_{rr} = -H_{zz}$*); one independent trace-free diagonal mode is absent. For a different orthonormal frame with columns* E, *the physical matrix is* EHE^{T}*; its componentwise maximum need not equal* m.

Proof. Substitute the seven vectors (12) and their coefficient perturbations (13) into the six independent entries of G. The diagonal entries are $-3D/2, -D/2, D/2$; the off-diagonal entries in $(r\theta, rz, \theta z)$ order are (C, A, B). Thus $\operatorname{tr} G = -3D/2$, and the scalar gauge addition $DI/2$ removes its trace. The maximum-entry identity then follows directly from the displayed matrix. Conjugation gives the last claim. □

Proposition 4.3 (The exact original seven-coefficient cone)**.** *For any* $\tau \geq 0$, *all seven constraints* $|\delta c_i| \leq \tau$ *of* (13) *hold if and only if*

$$4|D| + 2|C| \leq \tau, \qquad 4|D| + 2|A - 4B| \leq \tau, \qquad 8|B| \leq \tau. \tag{17}$$

The sufficient bound $m \leq \tau/14$ *therefore need not hold for a valid coefficient perturbation: at* $(A, B, C, D) = (\tau, \tau/8, 0, 0)$, *the exact largest coefficient perturbation and* m *both equal* τ.

Proof. For real x, y the identity $\max(|x + y|, |x - y|) = |x| + |y|$ is exact. Apply it to the paired coefficients $(\delta c_2, \delta c_3)$, $(\delta c_4, \delta c_5)$, and $(\delta c_6, \delta c_7)$. The first pair contributes $4|D| + 2|C|$, the second $4|D| + 2|A - 4B|$, and the third $8|B|$. The remaining coefficient $|\delta c_1| = 3|D|/2$ is dominated by the first pair. Substitution verifies the equality case. □

This identifies a previously implicit assumption: the bound (15) is also a physical entrywise bound for the trace-free perturbation *in the specified cylindrical frame*. The same conclusion requires a verified change of frame when H8 rays are tilted; pointwise orthogonal conjugation alone does not control the additional derivatives of a spatially varying frame.

5 A source-matched weighted dual obstruction

The coefficient condition (15) involves four trace-free coordinates, not arbitrary physical tensor components. To test a particularly strong candidate estimate, let

$$P(r)=\begin{cases}(r-1)^2(2-r)^2, & 1\le r\le 2,\\ 0, & \text{otherwise},\end{cases}\qquad \eta(z)=\begin{cases}(1-z^2)^3, & |z|\le 1,\\ 0, & \text{otherwise}.\end{cases} \tag{18}$$

Set $\rho(r,z)=P(r)\eta(z)\ge 0$ and choose the meridional source $f=P(r)\eta'(z)e_z$. This is source-matched in the literal sense that $S=\rho\, e_z\otimes e_z$ is a compact symmetric inverse: $\operatorname{div} S=f$. Also $\int r f_z\,\mathrm{d}r\,\mathrm{d}z=0$, so the force compatibility condition of Theorem 2.1 holds.

Proposition 5.1 (A pressure-independent stress bound)**.** *Let $\beta=15971/1720600$. No smooth compact symmetric tensor H and smooth scalar pressure q satisfy both*

$$\begin{aligned}\operatorname{div} H+\nabla q&=f,\\ |H_{ij}|&\le\beta\rho\quad(i,j\in\{r,\theta,z\}),\end{aligned} \tag{19}$$

where components refer to the orthonormal cylindrical frame*. The contradiction also persists for suitable smooth compact nonnegative approximations to the profiles in* (18)*.*

Proof. Construct an axisymmetric vector field with components

$$v_r=-rP\eta'',\qquad v_\theta=0,\qquad v_z=(2P+rP')\eta'. \tag{20}$$

It is divergence-free because $r^{-1}\partial_r(rv_r)+\partial_z v_z=0$. The zero-extended profiles give a compactly supported Lipschitz v: $P=P'=0$ at $r=1,2$, and $\eta'=\eta''=0$ at $z=\pm1$. Thus integration by parts against v is legitimate, by approximation if necessary, and the pressure term vanishes. With the angular factor 2π suppressed, the source pairing is

$$\begin{aligned}L&=\int_1^2\int_{-1}^1 r f_z v_z\,\mathrm{d}z\,\mathrm{d}r\\ &=\left(\int_1^2 rP^2\,\mathrm{d}r\right)\left(\int_{-1}^1(\eta')^2\,\mathrm{d}z\right)=\frac{1}{420}\,\frac{1024}{385}=\frac{256}{40425}.\end{aligned} \tag{21}$$

Here the middle equality follows by integrating r^2PP' by parts. The physical cylindrical gradient of v includes its hoop entry v_r/r. Its entrywise absolute sum is bounded by

$$|\nabla v|_1\le(4P+2r|P'|)|\eta''|+rP|\eta'''|+(3|P'|+r|P''|)|\eta'|. \tag{22}$$

The elementary exact radial integrals are $\int_1^2 P=1/30$, $\int_1^2 P^2=1/630$, $\int_1^2 rP^2=1/420$, and $\int_1^2 P|P'|=1/256$. On their supports, $r\le 2$, $|P''|\le 2$, $|\eta''|\le 6$ and $|\eta'''|\le 72|z|+120|z|^3$. Direct integration gives

$$\int_{-1}^1\eta=\frac{32}{35},\quad \int_{-1}^1\eta|\eta'|=1,\quad \int_{-1}^1\eta|z|=\frac14,\quad \int_{-1}^1\eta|z|^3=\frac{1}{20}.$$

Consequently, writing $U = \int_1^2 \int_{-1}^1 r\rho|\nabla v|_1 \,\mathrm{d}z\,\mathrm{d}r$, one obtains the rigorous upper bound

$$U \le \left(\frac{4}{420} + \frac{2}{64}\right)\frac{192}{35} + \frac{2}{315}\,24 + \frac{3}{128} + \frac{4}{15} = \frac{313361}{470400}. \tag{23}$$

But (19) would imply $L \le \beta U$ by the virtual-work identity $\int f \cdot v = -\int H : \nabla v$, whereas

$$L - \beta\,\frac{313361}{470400} = \frac{443015653}{2967690880000} > 0, \tag{24}$$

a contradiction. Nonnegative convolution approximations of P and η, supported away from the axis, converge in the requisite derivative integral norms. The strict gap survives for sufficiently small smoothing parameters and proves the smooth-profile assertion. □

Corollary 5.2 (Failure of the sufficient four-field envelope)**.** *For the source f and envelope ρ in (18), there is no fixed-frame four-coordinate stress $H(A,B,C,D)$ of (16) satisfying $\operatorname{div} H + \nabla q = f$ and the sufficient bound (15), even if the pressure is freely chosen.*

Proof. Proposition 4.2 transfers (15) to the entrywise physical cap (19). This contradicts Proposition 5.1. □

The preceding argument excludes the *stronger sufficient* four-field estimate. It cannot alone exclude the original absolute coefficient budget $\max_i |\delta c_i| \le (15971/122900)\rho$: if one simply inserts that larger constant into the conservative gradient bound (23), then

$$\frac{15971}{122900}\frac{313361}{470400} - L = \frac{51024386641}{635933760000} > 0. \tag{25}$$

The sign of this bound is inconclusive, not evidence of existence. The exact cone, however, has a smaller sharp dual support than that componentwise comparison.

Proposition 5.3 (Sharp dual of the original absolute coefficient cone)**.** *For $\tau \ge 0$ let $\mathcal{K}_\tau$ be the four-coordinate set defined by (17). For arbitrary $a,b,c,d \in \mathbb{R}$,*

$$\sup_{(A,B,C,D)\in\mathcal{K}_\tau} (aA + bB + cC + dD) = \tau\left[\max\left(\frac{|a|+|c|}{2}, \frac{|d|}{4}\right) + \frac{|4a+b|}{8}\right]. \tag{26}$$

The upper bound in (26) is proved in Lean for the actual seven coefficient constraints.

Proof. Put $x = A - 4B$. By (17), for fixed $s = |D| \in [0, \tau/4]$, the variables x, C, B are independent and satisfy

$$|x|, |C| \le \frac{\tau - 4s}{2}, \qquad |B| \le \frac{\tau}{8}.$$

The objective becomes $ax + (4a+b)B + cC + dD$. Its maximum for this s is

$$\frac{\tau - 4s}{2}(|a| + |c|) + \frac{\tau}{8}|4a+b| + s|d|.$$

This expression is affine in s, so its maximum occurs at $s = 0$ or $s = \tau/4$, yielding (26). Each bound is attained by choosing the signs of x, C, B, D to match their respective linear coefficients. Thus the value is sharp. □

Proposition 5.4 (An exact original-budget obstruction for the manufactured source)**.** *Let f, ρ be given by (18), and set $\gamma = 15971/122900$. No locally integrable symmetric tensor of the fixed-frame form $H(A,B,C,D)$ in (16), together with a locally integrable scalar pressure q, can satisfy in the distributional sense*

$$\operatorname{div} H + \nabla q = f, \qquad |\delta c_i(A,B,C,D)| \le \gamma\rho \quad (1 \le i \le 7).$$

Proof. Use the divergence-free meridional field v from (20). Its nonzero coupling coordinates to H are

$$a(r,z) = \partial_z v_r + \partial_r v_z = -rP\eta''' + (3P' + rP'')\eta',$$
$$d(r,z) = \partial_z v_z - \partial_r v_r = (3P + 2rP')\eta''.$$

Indeed, $H : \nabla v = Aa + Dd$; the B and C components vanish because v is meridional and independent of θ. By Proposition 5.3, integration by parts and $\rho \geq 0$ would force

$$L \leq \gamma S, \quad S = \int_1^2 \int_{-1}^1 rP(r)\eta(z) \max\left(|a|, \frac{|a|}{2} + \frac{|d|}{4}\right) \mathrm{d}z\, \mathrm{d}r. \tag{27}$$

There is an exact rational upper bound

$$S \leq \frac{5838537817787225523484624248207}{150323855360000000000000000000000} < \frac{1}{25}. \tag{28}$$

For completeness, partition $[1,2] \times [-1,1]$ into 40^2 rational rectangles. On each rectangle express the separated polynomials a and d in a common-degree bivariate Bernstein basis. The convex-hull property bounds $|a|$ and $|d|$ by the largest absolute Bernstein coefficient. Multiply $\max(a_{ij}, a_{ij}/2 + d_{ij}/4)$ by the *exact* nonnegative cell mass $\int rP\,\mathrm{d}r \int \eta\,\mathrm{d}z$, and sum the 1600 rational terms. This yields the first inequality in (28); the second is an integer comparison. The accompanying pure-rational program reproduces each coefficient and sum. Consequently

$$L - \gamma S > \frac{256}{40425} - \frac{15971}{122900 \cdot 25} = \frac{5637293}{4968232500} > 0,$$

contradicting (27). The pressure term vanishes by solenoidality of v. The zero-extended test field is compactly supported and Lipschitz, so smooth divergence-free mollifications justify the same integration-by-parts identity for locally integrable H and q. □

This obstruction concerns the *absolute correction budget* $|\delta c_i| \leq \gamma\rho$ in one fixed frame. It does not rule out every nonnegative seven-ray stress: the one-sided requirement $c_i^{(0)}\rho + \delta c_i \geq \rho/100$ admits perturbations outside that sufficient absolute budget. We now compute the exact support of this larger admissible set rather than extending the obstruction by analogy.

Proposition 5.5 (Sharp support of the one-sided seven-ray cone)**.** *Let $\rho \geq 0$ and let $\mathcal{P}_\rho$ contain precisely those (A,B,C,D) for which all seven original coefficients satisfy $c_i^{(0)}\rho + \delta c_i(A,B,C,D) \geq \rho/100$. Define*

$$g = \frac{15971}{122900}, \quad h = \frac{22371}{122900}, \quad x = A - 4B.$$

Then $\mathcal{P}_\rho$ is described exactly by

$$-\frac{g\rho}{4} \leq D \leq \frac{(g+h)\rho}{8}, \qquad |B| \leq \frac{g\rho}{8},$$
$$|x| \leq \frac{g\rho}{2} + 2D, \qquad \left|C - \frac{(h-g)\rho}{4}\right| \leq \frac{(g+h)\rho}{4} - 2D. \tag{29}$$

For arbitrary dual coefficients $(a,b,c,d) \in \mathbb{R}^4$ its support is

$$\sigma_\rho^+(a,b,c,d) = \rho\Bigg[\frac{h-g}{4}c + \frac{g}{8}|4a+b| + \max\left\{\frac{3g+h}{4}|c| - \frac{g}{4}d,\ \frac{3g+h}{4}|a| + \frac{g+h}{8}d\right\}\Bigg]. \tag{30}$$

The universal upper bound in (30) is kernel-checked in Lean under the original seven coefficient floors.

Proof. The six ordinary ray centres have available negative correction $-g\rho$ and the third ray has $-h\rho$. The paired fourth and fifth coefficient inequalities yield the bound on $|x|$; the sixth and seventh yield the bound on $|B|$. The second and third give the C interval, whose nonnegative half-width forces the upper bound on D. The half-width for x forces the lower bound on D. The remaining first-ray inequality $-3D/2 \geq -g\rho$ is then redundant, since $(g+h)/8 \leq 2g/3$. This proves the exact description, including when $\rho = 0$.

The objective is $ax + (4a+b)B + cC + dD$. For each fixed D, the independent intervals for x, B, C give a maximum equal to

$$\frac{h-g}{4}\rho c + \frac{g\rho}{8}|4a+b| + \left(\frac{g\rho}{2} + 2D\right)|a| + \left(\frac{(g+h)\rho}{4} - 2D\right)|c| + Dd.$$

This is affine in D. Substitution of its two endpoints gives (30). Signs of x, B and the displacement of C attain these extrema, establishing sharpness. □

Proposition 5.6 (The manufactured dual test is inconclusive for the one-sided cone)**.** *For the fixed diagnostic source and divergence-free test field in Proposition 5.4, let a, d be the coupling expressions there, and define*

$$J = \int_1^2 \int_{-1}^1 rP(r)\eta(z)\,\sigma_1^+(-a(r,z), 0, 0, -d(r,z))\,\mathrm{d}z\,\mathrm{d}r.$$

An exact rational partition certificate proves

$$J \geq \frac{24989775099966161649388574430033054 1}{38797083829862400000000000000000000000} > \frac{256}{40425} + \frac{1}{10000}. \tag{31}$$

Consequently the necessary dual inequality $L \leq J$ is not contradicted by this test field under the one-sided cone. This asserts neither the existence of a stress nor the absence of an obstruction from a different test field.

Proof. The explicit support (30), evaluated on $(-a, 0, 0, -d)$, is the maximum of the four signed polynomial actions

$$\frac{ga}{2} + \frac{gd}{4}, \quad -\frac{ga}{2} + \frac{gd}{4}, \quad \left(\frac{g}{2} + \frac{3g+h}{4}\right)a - \frac{g+h}{8}d, \quad -\left(\frac{g}{2} + \frac{3g+h}{4}\right)a - \frac{g+h}{8}d.$$

Divide the rectangle into 40×40 rational subrectangles. On each cell, the integral of the pointwise maximum is at least the maximum of the four *exact signed polynomial integrals* against the nonnegative weight $rP\eta$. Their rational sum gives the first inequality in (31). The second is an exact integer comparison. For independent error detection, the accompanying program also bounds the same integral from above with signed bivariate Bernstein coefficients. The pressure contribution vanishes because the test field is divergence free. □

The contrast is mathematically substantive: the absolute-budget obstruction is strict, whereas the exact one-sided cone is wider and this particular necessary inequality has positive slack. A primal positive-cone construction would additionally require a compact symmetric-divergence inverse with all support and derivative bounds. In addition, $f = P\eta' e_z$ is a manufactured diagnostic source, *not* the authenticated H8 residual $g - \mathbb{P}\,\mathrm{div}(\rho R_*)$, where $\mathbb{P}$ denotes the divergence-free Leray projection; the source profiles and potentially rotating physical frame required to test that residual have not been provided in the checked project. No statement here excludes a more flexible source-matched H8 inverse or decides the global unforced problem.

Proposition 5.7 (Physical-frame pullback of the one-sided dual)**.** *Let $E(x)$ be a pointwise orthogonal 3×3 source-to-physical frame, written in Cartesian components, and set $K(x) = E(x)H(A, B, C, D)(x)E(x)^{\mathsf{T}}$ with H from (16). For any physical gradient $G = \nabla v$, put $M = E^{\mathsf{T}}GE$ and define*

$$(a_E, b_E, c_E, d_E) = (M_{rz} + M_{zr},\ M_{\theta z} + M_{z\theta},\ M_{r\theta} + M_{\theta r},\ M_{zz} - M_{rr}). \tag{32}$$

Then

$$K : G = Aa_E + Bb_E + Cc_E + Dd_E. \tag{33}$$

If every original seven-ray coefficient is at least $\rho/100$ for $\rho \geq 0$, then the pointwise necessary bound is

$$-K : G \leq \sigma^+_\rho(-a_E, -b_E, -c_E, -d_E), \tag{34}$$

where σ^+ is given exactly by (30). Both (33) and (34) are kernel-checked in Lean as pointwise finite-matrix results; the algebra does not require E to be constant.

Suppose additionally that K and q solve $\operatorname{div} K + \nabla q = f$ distributionally on a periodic cell or a region where the following integration by parts is valid. For every smooth periodic or compactly supported solenoidal v,

$$\int f \cdot v \, \mathrm{d}x \leq \int \sigma^+_{\rho(x)}(-a_E(x), -b_E(x), -c_E(x), -d_E(x)) \, \mathrm{d}x. \tag{35}$$

Proof. The Frobenius identity $(EHE^{\mathsf{T}}) : G = H : (E^{\mathsf{T}}GE)$ follows by rearranging finite sums over the three matrix indices. Inserting the nine entries of H gives (33). Apply Proposition 5.5 to the negative of this dual covector to obtain (34). Finally $\int f \cdot v = -\int K : \nabla v$ because the pressure term pairs to zero with a solenoidal v, proving (35). □

The frame transfer changes the dual coordinates even when E is constant. For example, take $G = \operatorname{diag}(1, 2, 3)$ and the orthogonal meridional rotation

$$E = \begin{pmatrix} 3/5 & 0 & 4/5 \\ 0 & 1 & 0 \\ -4/5 & 0 & 3/5 \end{pmatrix}.$$

Formula (32) gives $(a_E, b_E, c_E, d_E) = (-48/25, 0, 0, -14/25)$, whereas the frozen frame gives $(0, 0, 0, 2)$. Thus a fixed-frame numerical obstruction cannot be transferred to a varying frame without recomputing its dual integral.

For a differentiable frame, the analytic divergence additionally contains the connection terms

$$\partial_j K_{ij} = E_{ia}E_{jb}\partial_j H_{ab} + (\partial_j E_{ia})H_{ab}E_{jb} + E_{ia}H_{ab}(\partial_j E_{jb}), \tag{36}$$

with repeated indices summed. Orthogonality ensures that a source-frame scalar multiple of the identity remains a physical pressure gauge, even if E varies. Equation (36) shows why the pointwise dual theorem alone does not control the weighted derivatives of a source-matched inverse. The authenticated functions g, ρ, R_* and E, together with their support and derivative budgets, remain required to evaluate (35) on the genuine H8 residual.

Proposition 5.8 (Exact pressure-compensated annular H8 gate)**.** *Fix a time and a cylindrical annulus $a < r < b$, $c < z < d$. Let K be a smooth symmetric physical tensor, p a smooth scalar, and $f = \operatorname{div} K + \nabla p$. Denote the azimuthal average by $\langle \cdot \rangle_\theta$ and set*

$$S(r) = \int_c^d \langle K_{rz} \rangle_\theta \, \mathrm{d}z, \qquad F(r) = \int_c^d \langle f_z \rangle_\theta \, \mathrm{d}z,$$
$$q_{\mathrm{s}}(r) = \left[\langle K_{zz} + p \rangle_\theta \right]_{z=c}^{z=d}, \qquad Q(r) = \frac{1}{r} \int_a^r s \, q_{\mathrm{s}}(s) \, \mathrm{d}s. \tag{37}$$

Then the exact cylindrical identity is

$$J_f(r) := \frac{1}{r}\int_a^r sF(s)\,\mathrm{d}s = S(r) - \frac{a}{r}S(a) + Q(r). \tag{38}$$

Suppose $K = EH(A,B,C,D)E^{\mathsf{T}}$, where $E(r,\theta,z)$ is orthogonal and the seven H8 ray coefficients exceed $\rho/100$ for $\rho \geq 0$. Let G be the physical matrix having $G_{rz} = 1$ and all other entries zero, and write λ_E for the four coordinates of (32) applied to this G. Define the exact slice supports

$$\begin{aligned}\mathcal{B}_{\pm}(r) &= \int_c^d \left\langle \sigma^{+}_{\rho(r,\theta,z)}(\pm\lambda_E(r,\theta,z)) \right\rangle_\theta \mathrm{d}z,\\ \mathcal{B}_{\mathrm{abs}}(r) &= \int_c^d \left\langle \sigma_{g\rho(r,\theta,z)}(\lambda_E(r,\theta,z)) \right\rangle_\theta \mathrm{d}z, \qquad g = \frac{15971}{122900}.\end{aligned} \tag{39}$$

The original absolute coefficient budget implies

$$\left| J_f(r) + \frac{a}{r}S(a) - Q(r) \right| \leq \mathcal{B}_{\mathrm{abs}}(r), \tag{40}$$

while the larger one-sided positive-coefficient cone requires only

$$-\mathcal{B}_-(r) \leq J_f(r) + \frac{a}{r}S(a) - Q(r) \leq \mathcal{B}_+(r). \tag{41}$$

These bounds hold for a spatially varying orthogonal frame; they are necessary conditions and do not construct a differential inverse.

Proof. The azimuthal mean removes the $r^{-1}\partial_\theta K_{\theta z}$ term in the cylindrical divergence. The axial component therefore obeys $F = S' + S/r + q_{\mathrm{s}}$, proving $(rS)' = r(F - q_{\mathrm{s}})$ and (38) by integration. At each spatial point $K_{rz} = K : G = \lambda_E \cdot (A,B,C,D)$. The sharp one-sided support formula (30) gives the upper and lower bounds on this quantity, and (26) gives the symmetric absolute bound. Azimuthal and axial integration followed by (38) proves both inequalities. □

The boundary primitive cannot be discarded in a local test. For example, on $1 \leq r \leq 2$, $0 \leq z \leq 1$, take $K = 0$ and $p = z$. Then $F = q_{\mathrm{s}} = 1$ and $S = 0$, whereas $J_f(r) = (r^2 - 1)/(2r)$ is nonzero. The zero stress satisfies either cone with $\rho = 0$. Thus $|J_f| \leq \mathcal{B}$ without the boundary correction would give a false obstruction. This example is local and does not assert compact pressure support.

For a finite exact check, let the frame be independent of z, θ on each radial slice and write $\bar{\rho}(r) = \int_c^d \rho(r,z)\,\mathrm{d}z$. Then the supports in (39) reduce to their respective finite-dimensional formulas at weight $\bar{\rho}$. For the rationally rotating frame

$$E(r) = \begin{pmatrix} \frac{1-t^2}{1+t^2} & 0 & \frac{2t}{1+t^2} \\ 0 & 1 & 0 \\ -\frac{2t}{1+t^2} & 0 & \frac{1-t^2}{1+t^2} \end{pmatrix}, \qquad t = r - 1,$$

the physical rz shear has pulled-back covector $\lambda_E = (-7/25, 0, 0, 24/25)$ at $r = 3/2$. Writing $h = 22371/122900$, exact evaluation gives

$$\mathcal{B}_+ = \frac{19h + 47g}{100}\bar{\rho}, \qquad \mathcal{B}_- = \frac{19g}{50}\bar{\rho}, \qquad \mathcal{B}_{\mathrm{abs}} = \frac{19g}{50}\bar{\rho}. \tag{42}$$

The asymmetry is real: replacing the original absolute budget by one-sided positivity can change a failed test into an inconclusive one. These rational point evaluations are not uniform estimates for a smooth shell, and no source authenticity follows from them.

Proposition 5.9 (Source-shaped meridional flux, including viscosity)**.** *Let $U = (U_r, U_\theta, U_z)$ be an axisymmetric divergence-free smooth velocity on the cylindrical slice. Consider finitely many poloidal stream functions*

$$\Psi_i(r,z) = K_i a_i(r) b(r) \Xi_i(z), \qquad P_{r,i} = -r^{-1}\partial_z \Psi_i, \quad P_{z,i} = r^{-1}\partial_r \Psi_i,$$

where K_i are constant and $\Xi_i = \Xi_i' = 0$ at both axial endpoints. Put $P = \sum_i P_i$ and allow any purely azimuthal toroidal field $T_\theta e_\theta$. Define the axial moments

$$I_i(r) = \int_c^d U_r(r,z)\Xi_i(z)\,\mathrm{d}z, \quad m_i = \int_c^d \Xi_i(z)\,\mathrm{d}z, \quad J_{ij} = \int_c^d \Xi_i'(z)\Xi_j(z)\,\mathrm{d}z.$$

Then the axial integral of the physical quadratic rz flux increment is

$$\begin{aligned}
\mathcal{C}(r) &= \int_c^d \left[(U+P+T)_r (U+P+T)_z - U_r U_z\right]\mathrm{d}z = \Gamma(r) b'(r) + \Delta(r) b(r) + Q_2(r) b(r)^2, \\
\Gamma &= \frac{1}{r}\sum_i K_i a_i I_i, \qquad \Delta = \frac{1}{r}\sum_i K_i \left[a_i' I_i - a_i\left(I_i' + \frac{I_i}{r}\right)\right], \\
Q_2 &= \frac{1}{r^2}\sum_{i<j} K_i K_j (a_j a_i' - a_i a_j') J_{ij}.
\end{aligned} \tag{43}$$

The contribution to the same flux from the canonical Newtonian viscous stress of P is exactly

$$\mathcal{V}_\nu(r) = -\nu \int_c^d (\partial_z P_r + \partial_r P_z)\,\mathrm{d}z = -\nu \sum_i K_i m_i \frac{d}{dr}\left(\frac{(a_i b)'}{r}\right). \tag{44}$$

In particular its coefficient of b'' is $-\nu r^{-1}\sum_i K_i m_i a_i$; it cannot be included in a $O(b^2)$ remainder merely by assigning a flat radial cutoff.

Proof. The mixed core–shaper product is $r^{-1}\sum_i K_i[(a_i b)' I_i - a_i b \int_c^d U_z \Xi_i'\,\mathrm{d}z]$. Axial integration by parts and $r^{-1}\partial_r(rU_r) + \partial_z U_z = 0$ give $\int U_z \Xi_i' = I_i' + I_i/r$, proving the linear terms. The self-products vanish because $\int \Xi_i' \Xi_i = 0$. For $i \neq j$ integration by parts gives $J_{ji} = -J_{ij}$; the paired cross-products reduce to $r^{-2} K_i K_j (a_j a_i' - a_i a_j') b^2 J_{ij}$. All rz components of a purely toroidal velocity vanish. Finally the integral of $\partial_z P_r$ vanishes by the endpoint conditions and $\int \partial_r P_z = \sum_i K_i m_i \partial_r((a_i b)'/r)$. This proves the viscous identity and its displayed b'' coefficient. □

For an exact local diagnostic take $U_r = -2r$, $U_z = 4z$, $c = -1$, $d = 1$, $b = (r-1)^3$, $\nu = 1$, and two shapers

$$(K_1, a_1, \Xi_1) = (1, 1, (1-z^2)^2), \qquad (K_2, a_2, \Xi_2) = (2, r, z(1-z^2)^2).$$

At $r = 2$, direct polynomial integration and the reduced formula independently give

$$\Gamma = -\frac{32}{15}, \quad \Delta = \frac{32}{15}, \quad Q_2 = \frac{64}{315}, \quad \mathcal{V}_1 = -\frac{12}{5}, \quad \mathcal{C} + \mathcal{V}_1 = -\frac{2036}{315}. \tag{45}$$

The polynomial functions illustrate the exact algebra on a local slice; their zero extensions are not C^∞ compact H8 cutoffs. Neither the still-free localization stresses nor the common pressure have been inserted into this number.

Proposition 5.10 (Flat-edge second-derivative obstruction)**.** *Under the hypotheses of Proposition 5.9, put* $M(r) = \sum_i K_i m_i a_i(r)$*. The shaper-induced kinetic and viscous stress-shear increment has the exact decomposition*

$$\begin{aligned} &\mathcal{C} + \mathcal{V}_\nu = \kappa_2 b'' + \kappa_1 b' + \kappa_0 b + \kappa_q b^2, \\ &\kappa_2 = -\frac{\nu M}{r}, \qquad \kappa_1 = \Gamma - \nu\left(\frac{2M'}{r} - \frac{M}{r^2}\right), \\ &\kappa_0 = \Delta - \nu\left(\frac{M''}{r} - \frac{M'}{r^2}\right), \qquad \kappa_q = Q_2. \end{aligned} \tag{46}$$

Let $r_0 > 0$ *and* $I = [r_0, r_0 + \delta]$*, and suppose these four coefficients are bounded on* I *with* $\inf_I |\kappa_2| > 0$*. If* $b \in C^2(I)$ *satisfies* $b(r_0) = b'(r_0) = 0$ *and another stress-shear contribution* R *obeys, for finite constants* $L, L' \geq 0$*,*

$$|R| \leq L(|b| + |b'| + b^2), \qquad |\mathcal{C} + \mathcal{V}_\nu + R| \leq L' b^2 \quad \text{on } I, \tag{47}$$

then $b \equiv 0$ *throughout* I*.*

Proof. Differentiating $M'b + Mb'$ divided by r in (44) yields $Mb''/r + (2M'/r - M/r^2)b' + (M''/r - M'/r^2)b$, which proves (46). Let $\eta = \inf_I |\kappa_2| > 0$ and $L_b = \sup_I |b| < \infty$. The two bounds in (47), boundedness of the other three coefficients and $b^2 \leq L_b|b|$ yield a constant $C < \infty$ such that

$$|b''(r)| \leq C(|b(r)| + |b'(r)|) \quad (r \in I).$$

By the initial conditions and the fundamental theorem of calculus, the continuous nonnegative function $w = |b| + |b'|$ satisfies $w(r) \leq (1 + C)\int_{r_0}^r w(s)\,\mathrm{d}s$. Gronwall's elementary integral inequality gives $w \equiv 0$, and therefore $b \equiv 0$. □

For the two profiles in (45), direct integration gives $m_1 = 16/15$ and $m_2 = 0$; hence $M = 16/15$ and

$$\mathcal{C} + \mathcal{V}_1 = -\frac{16}{15r}b'' + \left(-\frac{32}{15} + \frac{16}{15r^2}\right)b' + \frac{64}{15r}b + \frac{256}{315r^2}b^2. \tag{48}$$

At $r = 2$ and $b = (r-1)^3$ this again gives $-2036/315$. Since $\kappa_2 = -16/(15r)$ is nowhere zero on any annulus with $r > 0$ (indeed $|\kappa_2| \geq 8/15$ on $1 \leq r \leq 2$), Proposition 5.10 prohibits a nontrivial one-sided flat cutoff from satisfying an $O(b^2)$ budget for this *restricted* flux whenever its remaining terms obey the first inequality of (47).

In the H8 application, the two slice support functions in (39) yield such a target bound only if the relevant reference-centre-subtracted shear is precisely the quantity in (47) and both supports are $O(b^2)$ on the edge collar. Neither follows from the source architecture alone. Nontrivial compactness can instead require a vanishing effective b'' coefficient, a compensating transition or overlap term of second-jet size, a pressure boundary primitive with the requisite source matching, or a density budget larger than $O(b^2)$. The lemma does not rule out these possibilities and is not a contradiction for the authentic H8 residual.

The Case 006 source specifies a compact curl localization $u_b = \nabla \times (\chi A_c)$, separable poloidal and toroidal shell shapers, and a toroidal pressure radial derivative, but does not fix the smooth profiles, the complete four-patch interactions, or a common pressure representative. Hence no value of (38) has yet been certified for the complete H8 residual. The exact code records the transition, shaper, viscosity, overlap, reference-centre and axial boundary channels separately; omitting any named channel blocks evaluation. Supplying zero for a channel requires mathematical source justification, not merely successful execution of the code.

Proposition 5.11 (Quarter-turn localization and its nonlinear overlaps)**.** *Let $Q(x,y,z) = (-y,x,z)$ and χ be invariant under Q. On a simply connected Cartesian chart let A be a smooth vector potential, and denote the active push-forward by $Q_*v(x) = Qv(Q^{-1}x)$. Write*

$$v_j = Q_*^j(\nabla \times (\chi A)), \quad 0 \le j \le 3, \qquad u_b = \frac{1}{4}\sum_{j=0}^{3} v_j.$$

Then u_b is solenoidal and equals the curl of the normalized quarter-turn average of χA. Its full kinetic stress splits exactly into

$$u_b \otimes u_b = \frac{1}{16}\sum_{j=0}^{3} v_j \otimes v_j + \frac{1}{16}\sum_{\substack{0\le j,k\le 3\\ j\neq k}} v_j \otimes v_k. \tag{49}$$

For the material derivative, the same diagonal and ordered-cross decomposition holds with $v_j \otimes v_k$ replaced by $(v_j \cdot \nabla)v_k$. The twelve cross products do not generally vanish under fourfold symmetry.

Proof. Rigid rotations commute with curl, and the curl of an average is the average of the curls. Each v_j is divergence free. Expanding the product of the two sums for u_b proves (49); applying the Leibniz bilinearity of directional differentiation proves the derivative formula. The normalization $1/16$ follows from two factors of $1/4$ and cannot be dropped. □

For example, on a local chart take $\chi = 1$ and $A = (-yz, xz, 0)$. Every rotated field is $v_j = (-x,-y,2z)$, so in cylindrical notation $u_r = -r$, $u_z = 2z$, and $u_r u_z = -2rz$. The diagonal terms in (49) contribute $-rz/2$ to this shear, whereas the ordered overlaps contribute $-3rz/2$. Thus three quarters of the core shear is carried by the overlap partition even for an exactly fourfold-equivariant field. This local polynomial example does not claim a compact source profile; spatial disjointness, if asserted for different carrier supports, must be checked separately.

Proposition 5.12 (Shared-envelope second jet and core circulation)**.** *Let $r_0 < r < r_1$ be a cylindrical annulus separated from the axis, with angular coordinate θ and fixed axial endpoints $z_- < z_+$. Let $\mathcal{A} = (\mathcal{A}_r, \mathcal{A}_\theta, \mathcal{A}_z)$ be a smooth, 2π-periodic vector potential in the cylindrical orthonormal frame (possibly only C_4-equivariant), and let $U_c = \nabla \times \mathcal{A}$. Suppose the localized core and the poloidal stream functions of Proposition 5.9 share one radial envelope $b(r)$, so their total meridional correction is obtained from*

$$\nabla \times \left[b(r)\left(\mathcal{A} + \sum_i \frac{K_i a_i(r)\Xi_i(z)}{r} e_\theta \right)\right].$$

An arbitrary smooth background independent of b may be included. Denote the full angular mean by $\langle\cdot\rangle_\theta$, set

$$M(r) = \sum_i K_i m_i a_i(r), \qquad V(r) = \int_{z_-}^{z_+} \langle (U_c)_z(r,\theta,z)\rangle_\theta \, \mathrm{d}z,$$

and let $S_b(r)$ be the axial/angular integral of the increment in the physical shear $K_{rz} = u_r u_z - \nu(\partial_z u_r + \partial_r u_z)$. The coefficient of $b''(r)$ in S_b is exactly

$$\kappa_{2,\mathrm{eff}}(r) = -\nu\left[\int_{z_-}^{z_+} \langle \mathcal{A}_\theta(r,\theta,z)\rangle_\theta \, \mathrm{d}z + \frac{M(r)}{r}\right]. \tag{50}$$

All the remaining terms in S_b are expressions in b, b' with smooth bounded coefficients on compact subannuli. In particular, the quadratic overlaps among the four velocities contain no b'' term in this physical shear *channel.*

For any reference radius $r_* \in [r_0, r_1)$, put

$$H_* := r_* \int_{z_-}^{z_+} \langle \mathcal{A}_\theta(r_*, \theta, z) \rangle_\theta \, \mathrm{d}z.$$

Then (50) has the potential-free expression

$$\kappa_{2,\mathrm{eff}}(r) = -\frac{\nu}{r}\left[H_* + \int_{r_*}^{r} sV(s)\,\mathrm{d}s + M(r)\right]. \tag{51}$$

Consequently, identical *second-jet cancellation on an interval requires and is implied by a single anchor equality $H_* + \int_{r_*}^{r_a} sV(s)\,\mathrm{d}s + M(r_a) = 0$ and the differential identity*

$$rV(r) + M'(r) = 0 \tag{52}$$

throughout that interval. If the potential extends smoothly across the axis on complete disks, one may set $r_* = 0$ and $H_* = 0$.

Proof. Writing $A_\theta^{\mathrm{tot}} = \mathcal{A}_\theta + \sum_i K_i a_i \Xi_i / r$, the cylindrical curl gives

$$u_r - U_r = b\left(\frac{1}{r}\partial_\theta \mathcal{A}_z - \partial_z A_\theta^{\mathrm{tot}}\right), \qquad u_z - U_z = b\left(\frac{1}{r}\partial_r(rA_\theta^{\mathrm{tot}}) - \frac{1}{r}\partial_\theta \mathcal{A}_r\right) + b' A_\theta^{\mathrm{tot}}.$$

The quadratic product $u_r u_z$ therefore uses at most b and b'. In the viscous shear only $\partial_r(u_z - U_z)$ produces b'', with coefficient $-\nu A_\theta^{\mathrm{tot}}$. Angular/axial integration proves (50). Meanwhile

$$(U_c)_z = \frac{1}{r}\partial_r(r\mathcal{A}_\theta) - \frac{1}{r}\partial_\theta \mathcal{A}_r.$$

The full angular average annihilates the last derivative; hence $\partial_r(r\int\langle \mathcal{A}_\theta\rangle_\theta \,\mathrm{d}z) = rV(r)$. Integrating from r_* proves (51). Differentiating its bracket proves the stated characterization of identical cancellation. Smooth axis extension forces $r\langle \mathcal{A}_\theta\rangle_\theta \to 0$ as $r \downarrow 0$. □

The annular reference circulation matters. For the single profile $(K, a, \Xi) = (1, 1, (1 - z^2)^2)$ on $[-1, 1]$, one has $M = 16/15$. Choosing the local, C_4-invariant azimuthal potential $\mathcal{A}_\theta = -8/(15r)$ gives $\int_{-1}^{1} \mathcal{A}_\theta \,\mathrm{d}z = -16/(15r)$ and $\kappa_{2,\mathrm{eff}} \equiv 0$. Here $(U_c)_z = 0$ on the annulus, while $H_* = -16/15$ supplies the cancellation. This is an algebraic counterexample to deleting the circulation constant, not an admissible axis-spanning H8 core. If a smooth axis-spanning potential is prescribed, its core axial flux rather than a free annular circulation must supply the needed moment.

The proposition concerns the canonical physical rz stress. The pressure-compensated source relation (38) retains its independent axial jump; neither pressure nor an H8 inverse stress is reconstructed here. If the actual transition cutoff χ differs from the shaper envelope b, the core contributions cannot be assigned to the same b'' coefficient without an additional source identity. Thus the genuine H8 test still requires the core flux V, the reference circulation, the shared-cutoff decision, and the full source-matched cone density and frame.

The form of a compact cutoff may be fixed in a *test family* without identifying it with the source's unfrozen choice. Define

$$h(s) = \begin{cases} 0, & s \le 0, \\ \dfrac{\exp(-1/s)}{\exp(-1/s) + \exp(-1/(1-s))}, & 0 < s < 1, \\ 1, & s \ge 1, \end{cases} \qquad \chi_*(r) = h(r-1)h(4-r).$$

Then χ_* is smooth and nonanalytic, zero outside $[1,4]$, identically one on $[2,3]$, and equals $1/2$ at $r = 3/2, 7/2$. As a radial function on the surrounding chart it is C_4-invariant; the hole at $r \le 1$ makes its extension across the axis smooth. This provides a reproducible candidate for testing derivative estimates, not a determination of the H8 similarity-coordinate cutoff.

The source's toroidal pressure prescription likewise fixes only

$$\partial_r p_T = \frac{2u_\theta T_\theta + T_\theta^2}{r}. \tag{53}$$

For a chosen inner-radius normalization $p_T(a,z) = 0$, its representative is

$$p_T(r,z) = \int_a^r \frac{2u_\theta(s,z)T_\theta(s,z) + T_\theta(s,z)^2}{s}\,\mathrm{d}s.$$

For $a = 1$, $u_\theta = r$ and $T_\theta = rz(1-z)$ this gives

$$p_T = \frac{r^2-1}{2}(2z(1-z) + z^2(1-z)^2), \qquad [p_T]_{z=0}^{z=1} = 0.$$

By contrast $T_\theta = rz$ gives the nonzero pressure jump $[p_T]_{-1}^{1} = 2(r^2-1)$. The arbitrary radial integration trace $C(z)$ must be fixed by the remaining physical pressure equations; it is *not* an arbitrary gauge of the complete Navier–Stokes system.

To test all pieces in one local example, take $0 \le z \le 1$, $U_r = -r$, $U_z = 2z$, $U_\theta = r$, and the one-shaper data $K = a = 1$, $\Xi = z^2(1-z)^2$, $b = (r-1)^3$, $\nu = 1$, with the toroidal profile and pressure from the preceding zero-jump example. The C_4 core shear is partitioned into a diagonal $-r/4$ and ordered overlaps $-3r/4$; the shaper and viscous parts follow from Proposition 5.9. Directly integrating $K_{rz} = u_r u_z - (\partial_z u_r + \partial_r u_z)$ gives exactly

$$S(r) = \frac{-r^4 - 30r^3 - 2r + 3}{30r^2}, \qquad q_{\mathrm{s}}(r) = 4, \qquad J_f(r) = \frac{-r^4 + 30r^3 - 32r + 3}{30r^2}. \tag{54}$$

At $r = 3/2$ the exact values are $S = -63/40$ and $J_f = 91/120$. The independently computed axial source integral $F = (rS)'/r + q_{\mathrm{s}}$ reproduces J_f by integration from $a = 1$. The example uses a *polynomial local core and shaper*, not χ_*; it verifies the algebraic combination and avoidance of double-counted core terms. The smooth nonanalytic cutoff and the authentic Case 006 transition, frame, and reference-centre fields still require their own complete weighted calculation.

6 The physical residual and terminal speed

Define the viscosity-one residual

$$\mathcal{N}(u,p) = u_t + (u\cdot\nabla)u - \Delta u + \nabla p. \tag{55}$$

For smooth (u,p) and (w,q), direct expansion yields

$$\mathcal{N}(u+w, p+q) = \mathcal{N}(u,p) + \mathcal{N}(w,q) + (u\cdot\nabla)w + (w\cdot\nabla)u. \tag{56}$$

Both nonlinear cross terms are essential. Equation (56), with explicit differentiability assumptions in the upstream physical definitions, is checked in Lean by the declaration `UnforcedPatching.navierStokesResidual_add`.

Proposition 6.1 (Conditional terminal retention)**.** *Suppose that u has arbitrarily large speed at points with $t \uparrow 1$. If a correction w satisfies*

$$\|w(t,x)\| \le \alpha\|u(t,x)\| + C_0, \qquad 0 \le \alpha < 1, \quad C_0 \ge 0, \tag{57}$$

uniformly on $0 < t < 1$ and $x \in \mathbb{T}^3$, then $u + w$ also has arbitrarily large speed arbitrarily close to time one.

Proof. The reverse triangle inequality and (57) give

$$\|u + w\| \geq \|u\| - \|w\| \geq (1 - \alpha)\|u\| - C_0.$$

Given any target speed $M > 0$ and any terminal neighbourhood, choose a point therein where $\|u\| > (M + C_0)/(1 - \alpha)$. Then $\|u + w\| > M$ at that point. □

This last implication is kernel-checked by the Lean theorem

`UnforcedPatching.speedUnbounded_add_subleading`

It is deliberately conditional: no field u meeting the singularity premise, and no actual PDE correction w meeting (57), has been supplied by these lemmas.

7 What the formal development proves

The Lean project is pinned to Lean 4.34.0-rc2 and a fixed Mathlib revision. The checked physical interface states smoothness, three coordinate periods, zero forcing, nonzero initial data, finite fundamental-cell kinetic energy and endpoint speed divergence. Its existential proposition is named `UnforcedPatching.unforcedTargetStatement`; there is no proof term inhabiting that proposition.

The precise correspondence between the paper and the compiled `UnforcedPatching` declarations is as follows. In the second column, "missing" refers to mathematical constructions, not to unresolved Lean goals inside the named theorems.

Lean module	Checked scope and the missing construction
`Definitions`	Physical candidate predicates; no candidate witness.
`Residual`, `Correction`	Full residual addition and conditional cancellation; no constructed correcting field.
`Periodicity`, `PressureGauge`	Sum and time-only gauge identities; no periodic singular solution.
`ConeBudget`, `ConeOperator`, `SevenRayCenter`	Exact rational centre and sharp factor 14; no source-matched weighted positive stress.
`FourFieldPhysical`	Exact nine-entry ray-to-cylindrical tensor map modulo an isotropic gauge; entrywise four-coordinate bound in the frozen frame, not a spatially rotated PDE bridge.
`ExactCoefficientCone`, `SharpCoefficientDual`	Exact absolute-budget seven-coefficient polytope and sharp dual upper estimate; no authentic H8 source or spatial inverse.
`PositiveCoefficientCone`	Universal one-sided seven-ray positive-cone support bound; not a spatial stress inverse or a continuum integral formalization.
`RotatingFrameDual`	Exact source-to-physical matrix Frobenius pullback and pointwise rotated one-sided dual bound; no derivative-of-frame or PDE source matching theorem.
`CompactGauge`	One-variable gauge derivative; no compact tensor support theorem.
`MeridionalInverse`	Actual axial and radial interval-integral FTC identities, geometric hoop cancellation, and pointwise radial/axial cylindrical inverse; no Cartesian tensor-divergence bridge or compact support theorem.
`Retention`	Subleading corrections preserve terminal speed; neither required field constructed.

In particular, the residual-addition, correction-closure, seven-ray-floor, and terminal-retention declarations are all implications with explicit hypotheses. Their conjunction does not yield an inhabitant of `unforcedTargetStatement`. The printed axioms for the selected smoke-tested declarations are the standard logical axioms `propext`, `Classical.choice`, and `Quot.sound`.

In particular, `meridional_correction_divergence` checks both components of (3)–(4) for the hoop stress (9) and the actual interval-integral axial primitive (8), under stated local continuity and integrability assumptions. The separate declarations `radialMoment_relation` and `mixedCarrier_radial_balance` verify the antiderivative identity in (6) and its product carrier (7). The further theorem `sourceMatched_axialStress` checks the pointwise cylindrical source match $S_{zz} = \rho$, $f_z = \partial_z \rho$ used in Proposition 5.1. These results compile without proof admissions; selected axiom reports list only Lean's standard logical axioms. They remain pointwise statements about *defined cylindrical expressions*: the Cartesian tensor-divergence equivalence, global support and regularity construction, moment compatibility and weighted H8 estimate are not yet formalized in Lean. Thus the full restricted Theorem 2.1 is *partially*, not fully, formalized.

An independent exact-arithmetic test evaluates a nontrivial annular polynomial instance of the radial compatibility identity. For the uncompensated purely radial version, the curl obstruction at $(r, z) = (5/4, 1/2)$ equals $-5103/32768$ exactly; the test explicitly rejects the pressure-neutral interpretation in this case. The polynomial is used only for interior differentiation, not as an infinitely smooth zero-extended cutoff.

A second independent exact-rational test reproduces (21)–(24) from polynomial integrals and checks the complete cylindrical divergence of the dual test field. An additional exact reconstruction and Lean theorem prove the fixed-frame four-field map (16), its inverse coordinate recovery and its entrywise consequence; the test (25) records the failure of this *conservative gradient method* at the original coefficient budget. The sharper polytope method proves Proposition 5.4: a 40×40 Bernstein partition yields the exact rational upper bound (28). No executable test substitutes for the authentic H8 source and rotated-frame PDE estimates. The equivalence (17) is independently checked on exact rational samples and kernel-checked by `original_seven_coefficient_cone_exact`. The additional kernel theorem `original_cone_dual_upper` establishes the finite-dimensional upper bound in (26); exact rational maximizing witnesses independently verify sharpness. The Bernstein integral itself is a rational Python certificate, not a Lean theorem.

The separate one-sided cone is described by (29); its universal support upper bound is proved by the compiled Lean declaration `one_sided_positive_cone_dual_upper`. Independent exact rational feasible witnesses attain this value. A new 40×40 rational partition proves the lower bound (31), with a separate signed Bernstein upper enclosure. The integral certificate and its comparison with the source pairing are not formalized in Lean. The strict positivity of $J - L$ means only that this *particular* dual test supplies no one-sided positive-cone nonexistence certificate.

The further compiled declarations

```
rotating_fourField_frobenius
rotated_oneSided_dual_upper
```

establish (33) and (34) for every pointwise 3×3 frame matrix. Independent rational tests verify the rotated covector, equality of matrix contractions, an attainable one-sided dual support, and preservation of isotropic pressure gauges under orthogonal (but not arbitrary) frames. The spatial differential identity (36) and integrated inequality (35) are proved on paper, not yet in Lean; neither has been evaluated against the unavailable authentic H8 source.

The source code, machine-readable G0–G8 ledger, smoke tests and reproduction commands accompany the public integration branch [6]. A local pinned build has passed sixteen smoke

modules; at the time of preparing this edition, the corresponding GitHub-hosted Actions checks had not furnished a successful receipt for the latest commit. Local compilation does not replace hosted CI, semantic theorem auditing or independent scrutiny of the paper proof.

8 Remaining analytic requirements

The compact meridional inverse of Theorem 2.1 is a properly restricted solvability result. It does not resolve the weighted attachment problem because the desired source may have an azimuthal component, a noncompact Leray-projected tail, nontrivial pressure coupling, or incompatible moments. Even when compact inversion is possible, the output may leave the required seven-ray positive cone, especially as the support shrinks. The strict dual obstruction in Proposition 5.1, together with Corollary 5.2, excludes the sufficient four-coordinate estimate; Proposition 5.4 also excludes the original absolute coefficient correction budget for that same manufactured source. In contrast, Proposition 5.6 shows that the tested dual field has no obstruction for the larger one-sided cone; this leaves other dual tests and all primal construction questions open. The next mathematical question is whether the *authentic* source, perhaps using a tilted frame or a more flexible coefficient allocation, admits a weighted inverse satisfying the one-sided positive H8 ray conditions (or the absolute inequalities (17)) and all spatial derivative bounds. The sharp cone description enables a direct weighted dual optimization against the authentic source through (35), without replacing those inequalities by the factor-14 sufficient estimate. No numerical conclusion can be drawn from this inequality until the actual g, ρ, R_*, E are supplied and the frame-derivative terms in (36) are controlled.

A complete unforced singularity construction must still provide a single authenticated local profile, source-matched smooth shell data, a scale-sharp weighted symmetric inverse respecting the coefficient margin, a periodic nonlinear correction with a proved right-inverse/contraction scheme, the full physical zero-residual identity, and terminal retention for that actual correction. Each claim requires a further source-to-Lean semantic audit and a compiled theorem using the genuine PDE definitions.

Thus the present mathematical result is an exact compact stress construction and several independently checked necessary or conditional ingredients. The existence or nonexistence of a finite-time singular solution to (1) with smooth nonzero periodic initial data is *not decided by this manuscript*.

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